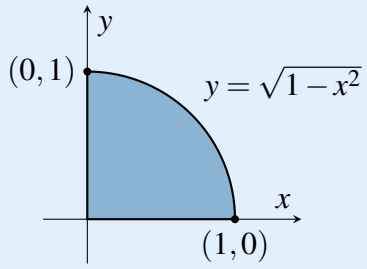
Name: _____ / 100

90 minutes maximum. No aids (book, calculator, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form. 100 points possible.

1. Consider the region in the first quadrant bounded by the curves $y = \sqrt{1-x^2}$, $x = 0$, and $y = 0$.

   (a) (4 points) Sketch the graph of the region. Be sure to clearly label any important points.

   Solution

   (b) (6 points) Write an integral that computes the volume of the solid obtained by rotating the region about the x-axis using the disk/washer method. **Do not compute the integral**.

   Solution

   Disks of radius $\sqrt{1-x^2}$ for $0 \le x \le 1$:

   $$\boxed{V = \int_0^1 \pi\left(\sqrt{1-x^2}\right)^2 dx}$$

   (c) (6 points) Write an integral that computes the volume of the solid obtained by rotating the region about the x-axis using the method of cylindrical shells. **Do not compute the integral**.

   Solution

   Cylindrical shells at height y have radius y and height $x = \sqrt{1-y^2}$, for $0 \le y \le 1$:

   $$\boxed{V = \int_0^1 2\pi y\sqrt{1-y^2}\, dy}$$

(d) (6 points) Compute one of the integrals from the previous two parts of this problem. You may choose which one to compute, but you must clearly indicate which one you have chosen.

Solution

Disk/washer integral from part (b):

$$V = \pi \int_0^1 (1-x^2)\,dx = \pi \left[x - \frac{x^3}{3}\right]_0^1 = \boxed{\frac{2\pi}{3}}.$$

Alternative (cylindrical shells, part (c)). With $u = 1-y^2$, $du = -2y\,dy$:

$$\int_0^1 2\pi y\sqrt{1-y^2}\,dy = -\pi \int_1^0 u^{1/2}\,du = \pi \left[\frac{2}{3}u^{3/2}\right]_0^1 = \frac{2\pi}{3}. \quad \checkmark$$

(e) (4 points) How could you have calculated the volume from the previous parts **without using calculus**?

Solution

The solid is a hemisphere of radius 1, so its volume is half that of a sphere:

$$V = \frac{1}{2}\cdot\frac{4}{3}\pi(1)^3 = \boxed{\frac{2\pi}{3}}.$$

2. (6 points) Set up, but **do not evaluate** an integral that computes the arc length of the curve $y = e^{3x}$ between the points $(0,1)$ and $\left(\frac{1}{3}, e\right)$.

Solution

$y' = 3e^{3x}$, so

$$\boxed{L = \int_0^{1/3} \sqrt{1+9e^{6x}}\,dx}$$

3. (10 points) A spring has a natural length of 2 meters. It requires 60 joules of work to stretch the spring to a length of 3 meters. How much would the spring stretch if you applied 15 joules of work? You may take Hooke's law ($F = kx$) as given, but must justify the rest of your answer with calculus.

Solution

Let x be the stretch beyond the natural length, so 3 m is $x = 1$, and let d be the stretch produced by 15 J:

$$60 = \int_0^1 kx\,dx = \frac{k}{2} \Rightarrow k = 120; \qquad 15 = \int_0^d 120x\,dx = 60d^2 \Rightarrow d^2 = \frac{1}{4}.$$

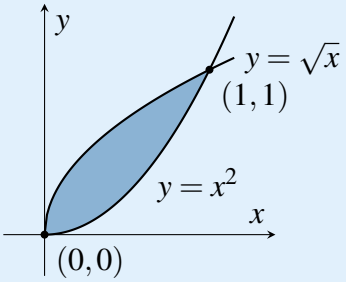
The spring stretches $\boxed{\frac{1}{2} \text{ meter}}$.

4. Consider the region bounded by $y = x^2$ and $y = \sqrt{x}$.

   (a) (4 points) Sketch the region. Be sure to clearly label any important points.

Solution

Intersections: $x^2 = \sqrt{x} \Rightarrow x^4 = x \Rightarrow x(x^3 - 1) = 0 \Rightarrow x = 0,\ 1$, giving $(0,0)$ and $(1,1)$.



   (b) (8 points) Set up expressions for the centroid (center of mass) of the region. **Do not evaluate** any integrals in your expressions.

Solution

On $0 \le x \le 1$ the top curve is $y = \sqrt{x}$ and the bottom is $y = x^2$.

$$\boxed{\bar{x} = \frac{\displaystyle\int_0^1 x\left(\sqrt{x} - x^2\right) dx}{\displaystyle\int_0^1 \left(\sqrt{x} - x^2\right) dx}} \qquad \boxed{\bar{y} = \frac{\displaystyle\int_0^1 \tfrac{1}{2}\left(\left(\sqrt{x}\right)^2 - \left(x^2\right)^2\right) dx}{\displaystyle\int_0^1 \left(\sqrt{x} - x^2\right) dx}}$$

5. Evaluate the following integrals and antiderivatives.

(a) (8 points) $\displaystyle\int_0^{\pi} x\sin(x)\,dx$

Solution

Integration by parts with $u = x$, $dv = \sin x\,dx$, $du = dx$, $v = -\cos x$:

$$\int_0^{\pi} x\sin x\,dx = \Big[-x\cos x\Big]_0^{\pi} + \int_0^{\pi} \cos x\,dx$$
$$= (-\pi\cos\pi - 0) + \Big[\sin x\Big]_0^{\pi} = \pi + 0 = \boxed{\pi}.$$

(b) (8 points) $\displaystyle\int x^2 e^{3x}\,dx$

Solution

Integration by parts with $u = x^2$, $dv = e^{3x}\,dx$, $du = 2x\,dx$, $v = \frac{1}{3}e^{3x}$:

$$\int x^2 e^{3x}\,dx = \frac{1}{3}x^2 e^{3x} - \frac{2}{3}\int xe^{3x}\,dx.$$

Again with $u = x$, $dv = e^{3x}\,dx$, $du = dx$, $v = \frac{1}{3}e^{3x}$:

$$\int xe^{3x}\,dx = \frac{1}{3}xe^{3x} - \frac{1}{3}\int e^{3x}\,dx = \frac{1}{3}xe^{3x} - \frac{1}{9}e^{3x}.$$

$$\boxed{\int x^2 e^{3x}\,dx = \frac{1}{3}x^2 e^{3x} - \frac{2}{9}xe^{3x} + \frac{2}{27}e^{3x} + C}$$

(c) (8 points) $\int \sin^2(x)\cos^2(x)\,dx$

Solution

Using $\sin^2\theta = \frac{1}{2}\big(1-\cos(2\theta)\big)$ and $\cos^2\theta = \frac{1}{2}\big(1+\cos(2\theta)\big)$:

$$\begin{aligned}\int \sin^2 x\cos^2 x\,dx &= \int \tfrac{1}{2}\big(1-\cos(2x)\big)\cdot\tfrac{1}{2}\big(1+\cos(2x)\big)\,dx = \frac{1}{4}\int \big(1-\cos^2(2x)\big)\,dx \\ &= \frac{1}{4}\int \Big(1-\tfrac{1}{2}\big(1+\cos(4x)\big)\Big)\,dx = \frac{1}{8}\int \big(1-\cos(4x)\big)\,dx \\ &= \boxed{\frac{x}{8}-\frac{\sin(4x)}{32}+C}\end{aligned}$$

(d) (6 points) $\int \sin^3(x)\cos^3(x)\,dx$

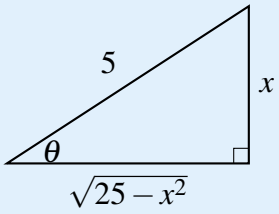
Solution

Write $\cos^3 x = (1-\sin^2 x)\cos x$ and let $u=\sin x$, $du=\cos x\,dx$:

$$\int \sin^3 x\,(1-\sin^2 x)\cos x\,dx = \int (u^3-u^5)\,du = \boxed{\frac{\sin^4 x}{4}-\frac{\sin^6 x}{6}+C}$$

(e) (8 points) $\displaystyle\int \frac{dx}{x^2\sqrt{25-x^2}}$

Solution



$\sin\theta = \dfrac{x}{5}$, so $x = 5\sin\theta$ and $dx = 5\cos\theta\, d\theta$.

$\cos\theta = \dfrac{\sqrt{25-x^2}}{5}$, so $\sqrt{25-x^2} = 5\cos\theta$.

$$\int \frac{5\cos\theta\, d\theta}{25\sin^2\theta \cdot 5\cos\theta} = \frac{1}{25}\int \csc^2\theta\, d\theta = -\frac{1}{25}\cot\theta + C.$$

From the triangle, $\cot\theta = \dfrac{\sqrt{25-x^2}}{x}$, so

$$\boxed{\int \frac{dx}{x^2\sqrt{25-x^2}} = -\frac{\sqrt{25-x^2}}{25x} + C}$$

(f) (8 points) $\displaystyle\int \frac{dx}{x^2-5x+6}$

Solution

$x^2-5x+6 = (x-2)(x-3)$, and

$$\frac{1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3} \quad\Longrightarrow\quad 1 = A(x-3) + B(x-2).$$

$x = 2$: $A = -1$. $\quad x = 3$: $B = 1$.

$$\int \left(\frac{1}{x-3} - \frac{1}{x-2}\right) dx = \boxed{\ln|x-3| - \ln|x-2| + C}$$

6. BONUS (5 points) Evaluate $\int e^{5x}\sin(7x)\,dx$

Solution

Let $I = \int e^{5x}\sin(7x)\,dx$. Integration by parts with $u = \sin(7x)$, $dv = e^{5x}\,dx$, $du = 7\cos(7x)\,dx$, $v = \frac{1}{5}e^{5x}$:

$$I = \frac{1}{5}e^{5x}\sin(7x) - \frac{7}{5}\int e^{5x}\cos(7x)\,dx.$$

Again with $u = \cos(7x)$, $dv = e^{5x}\,dx$, $du = -7\sin(7x)\,dx$, $v = \frac{1}{5}e^{5x}$:

$$\int e^{5x}\cos(7x)\,dx = \frac{1}{5}e^{5x}\cos(7x) + \frac{7}{5}I.$$

Substituting,

$$I = \frac{1}{5}e^{5x}\sin(7x) - \frac{7}{25}e^{5x}\cos(7x) - \frac{49}{25}I$$

$$\text{Solving for } I: \quad \frac{74}{25}I = \frac{1}{25}e^{5x}\big(5\sin(7x) - 7\cos(7x)\big).$$

$$\boxed{\int e^{5x}\sin(7x)\,dx = \frac{e^{5x}\big(5\sin(7x) - 7\cos(7x)\big)}{74} + C}$$